Roll No. ....

ODD END SEMESTER EXAMINATION DECEMBER 2025

Name of the Course: B. Tech. **Semester: I**
Name of the Paper: Engineering Mathematics for AI **Course Code: TMA 102**

Time: 3 Hours Maximum Marks: 100

Note:
- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries ten marks.

Q1	(20 marks)	
a)	Find rank of the given matrix by reducing it to Echelon form $A=\begin{bmatrix}1 & 3 & 2 & 5 & 1\\ 2 & 2 & -1 & 6 & 3\\ 1 & 1 & 2 & 3 & -1\\ 0 & 2 & 5 & 2 & -3\end{bmatrix}$	CO 1
b)	Find the matrix P and verify $P^{-1}AP = D$, where D is diagonal matrix for matrix $A = \begin{bmatrix}3 & -1 & 1\\ -1 & 5 & -1\\ 1 & -1 & 3\end{bmatrix}$	
c)	Verify the Cayley- Hamilton theorem and using Cayley-Hamilton theorem, find A^{-1} for the following matrix: $A = \begin{bmatrix}1 & -1 & 4\\ 3 & 2 & -1\\ 2 & 1 & -1\end{bmatrix}$	
Q2	(20 marks)	
a)	Let X = {1, 2, 3, 4}, and let R be a relation defined on X as: R = {(1, 1), (2, 1), (3, 1), (3, 2), (3, 4), (1, 4), (2, 3), (4, 4)} then draw the diagraph for R and find out M_R. Also find $RoR, RoRoR$ using M_R.	CO 2
b)	A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C++, 20 in Python, six in C++ and Java, one in Java and Python, five in C++ and Python, and just one programmer proficient	

	in all three languages above. Determine the number of computer programmers that are not proficient in any of these three languages.	
c)	Construct the truth table for the statement $[\sim p \wedge (\sim q \wedge r)] \vee (q \wedge r) \vee (p \wedge r)$	
Q3	(20 marks)	
a)	Show that: $$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 2\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}\right)$$ $$If\ \frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1\ is\ given.$$	CO 3
b)	Construct the Hessian for the function $-x^3 + 3xz + 2y - y^2 - 3z^2$ and find the extreme value(s) of the function.	
c)	The pressure P at any point in space is $P = 400xyz^2$. Find the highest pressure at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$.	
Q4	(20 marks)	
a)	Evaluate $\underset{x \to 0}{Lim} \left(\frac{sinx}{x}\right)^{\frac{1}{x^2}}$	
b)	If $y = \sin(msin^{-1}x)$, prove that $$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$$	COs 4
c)	Show that the functions:$u = x + y + z$,$v = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx$, $w = x^3 + y^3 + z^3 - 3xyz$ are functionally dependent. Find the relation between them.	
Q5	(20 marks)	
a)	Evaluate by changing the order of integration: $$\int_0^a \int_{\sqrt{ax}}^a \frac{y^2}{\sqrt{y^4 - a^2x^2}} dxdy$$	CO 5
b)	Prove that: $\int_0^{\pi/2} sin^m\theta cos^n\theta d\theta = \frac{\Gamma\frac{m+1}{2}\Gamma\frac{n+1}{2}}{2\Gamma\frac{m+n+2}{2}}$ and evaluate $\int_0^1 x^4(1-\sqrt{x})^5\, dx$	
c)	Evaluate $\iiint_R (x+y+z)\, dxdydz$, where R is the region bounded by $x = 0, y = 0, z = 0$ and $x + y + z = 1$	